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MIL-M-28787/534
30 March 1992

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TEMPERATURE SENSOR MODULE, KEY CODE EWB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic temperature sensor module.

1.2 Classification. This specification covers one class of module. Class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.774 failures per million operating hours for class I when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be EWB.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Units
				Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	MOhm
A2-A40	Isolation	II	X	10	INF	10	INF	MOhm
B1-B33	Continuity	III	X	0	1	0	1	Ohm
C	Vcc stress	IV	Q	---Module shall pass all remaining tests---				
D1,D2	Ibd	V	S	-0.239	0.339	-0.239	0.339	mA
D3	Ibd	V	S	-0.239	0.439	-0.239	0.439	mA
D4,D5	Ibd	V	S	-0.239	0.339	-0.239	0.339	mA
E1,E2	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-1.64	-0.46	-1.64	-0.46	mA
F2	Iih	VII	X	-2.83	-0.44	-2.83	-0.44	mA
F3	Iih	VII	X	-1.84	-0.46	-1.84	-0.46	mA
F4	Iih	VII	X	-2.24	-0.44	-2.24	-0.44	mA
F5,F6	Iih	VII	X	-1.84	-0.46	-1.84	-0.46	mA
F7	Iih	VII	X	-1.64	-0.46	-1.64	-0.46	mA
G1	Iil	VIII	X	-1.64	-0.46	-1.64	-0.46	mA
G2	Iil	VIII	X	-2.83	-0.44	-2.83	-0.44	mA
G3	Iil	VIII	X	-1.84	-0.46	-1.84	-0.46	mA
G4	Iil	VIII	X	-2.24	-0.44	-2.24	-0.44	mA
G5,G6	Iil	VIII	X	-1.84	-0.46	-1.84	-0.46	mA
G7	Iil	VIII	X	-1.64	-0.46	-1.64	-0.46	mA
H1,H2	Voh	IX	X	2.4	4.5	2.4	4.5	V
J1-J6	Vol	X	X	0.0	0.5	0.0	0.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Units
				Min	Max	Min	Max	
M	PSFC	XI	S	2.1	3.9	2.1	3.9	uF
N	Icc	XII	X	36.0	86.9	36.0	86.9	mA
P1,P2	Functional	XIII	X	See table XIII				
Q1,Q2	Dynamic	XIV	S	See table XIV				
Q3,Q4	Dynamic	XIV	Q	See table XIV				
R1	C1/o	XVI	Q	2	75	2	75	pF
R2	C1/o	XVI	Q	2	50	2	50	pF
R3	C1/o	XVI	Q	2	75	2	75	pF
R4	C1/o	XVI	Q	2	100	2	100	pF
R5,R6	C1/o	XVI	Q	2	75	2	75	pF
R7	C1/o	XVI	Q	2	50	2	50	pF
R8	C1/o	XVI	Q	8	75	8	75	pF
R9	C1/o	XVI	Q	6	75	6	75	pF
R10	C1/o	XVI	Q	4	75	4	75	pF
R11	C1/o	XVI	Q	2	75	2	75	pF
R12	C1/o	XVI	Q	2	100	2	100	pF
R13	C1/o	XVI	Q	2	125	2	125	pF
R14	C1/o	XVI	Q	2	75	2	75	pF
R15	C1/o	XVI	Q	2	100	2	100	pF
R16	C1/o	XVI	Q	2	75	2	75	pF
R17,R18	C1/o	XVI	Q	2	50	2	50	pF
R19-R21	C1/o	XVI	Q	2	75	2	75	pF
R22	C1/o	XVI	Q	2	100	2	100	pF
T1,T2	PSTR	XVII	Q	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/534-1 for class I. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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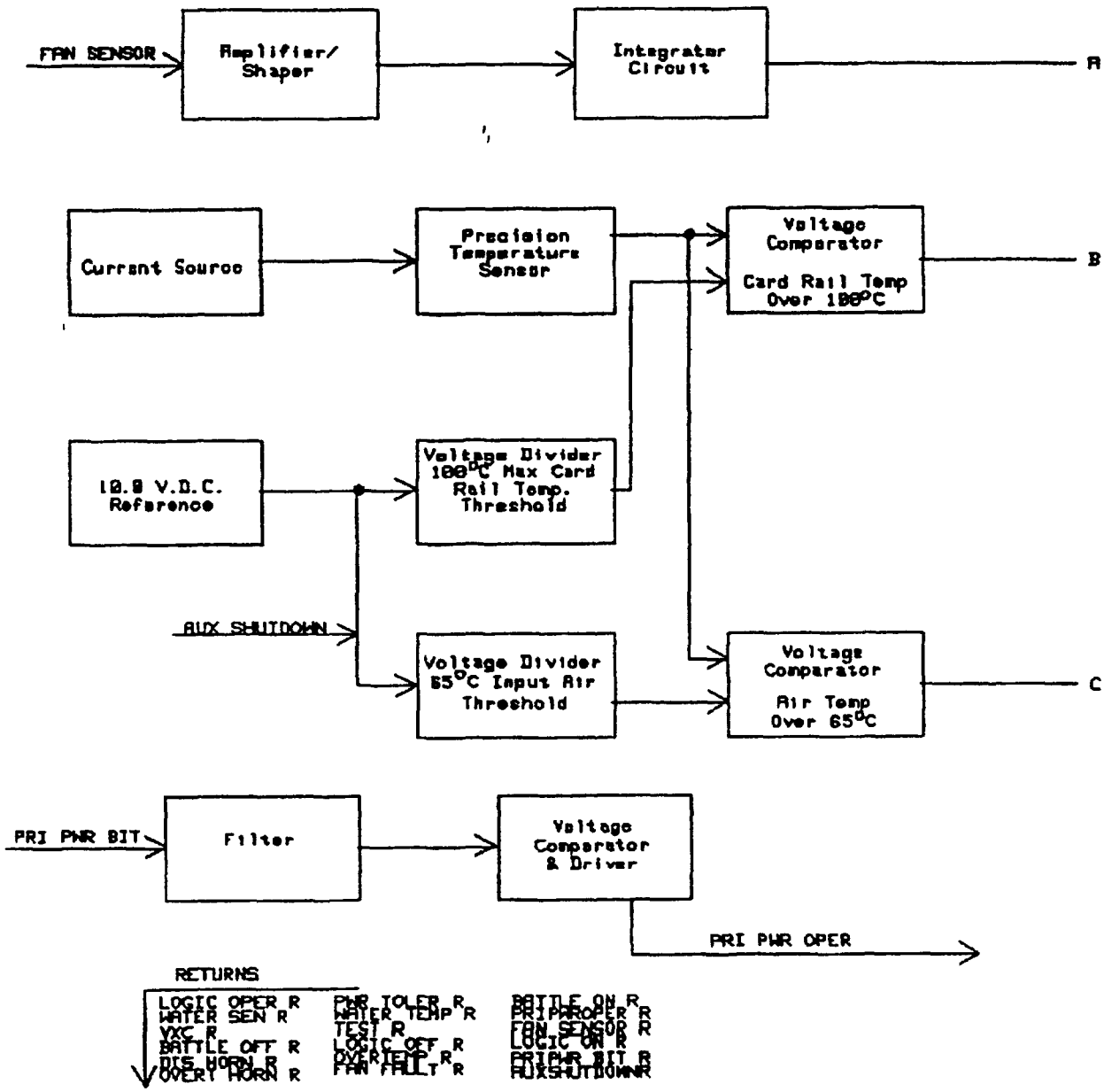


FIGURE 1. EWB functional block diagram.

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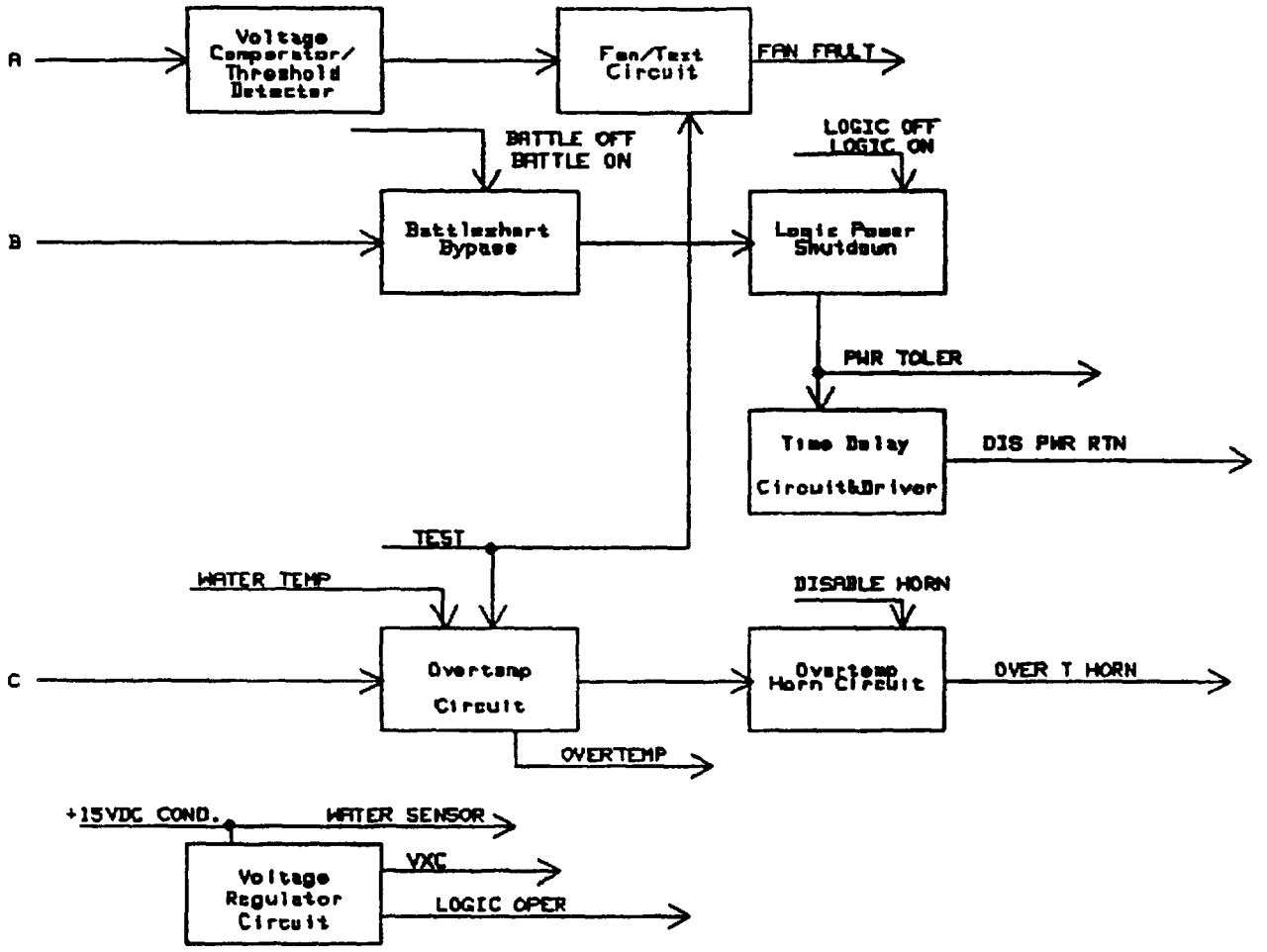


FIGURE 1. EWB functional block diagram - Continued.

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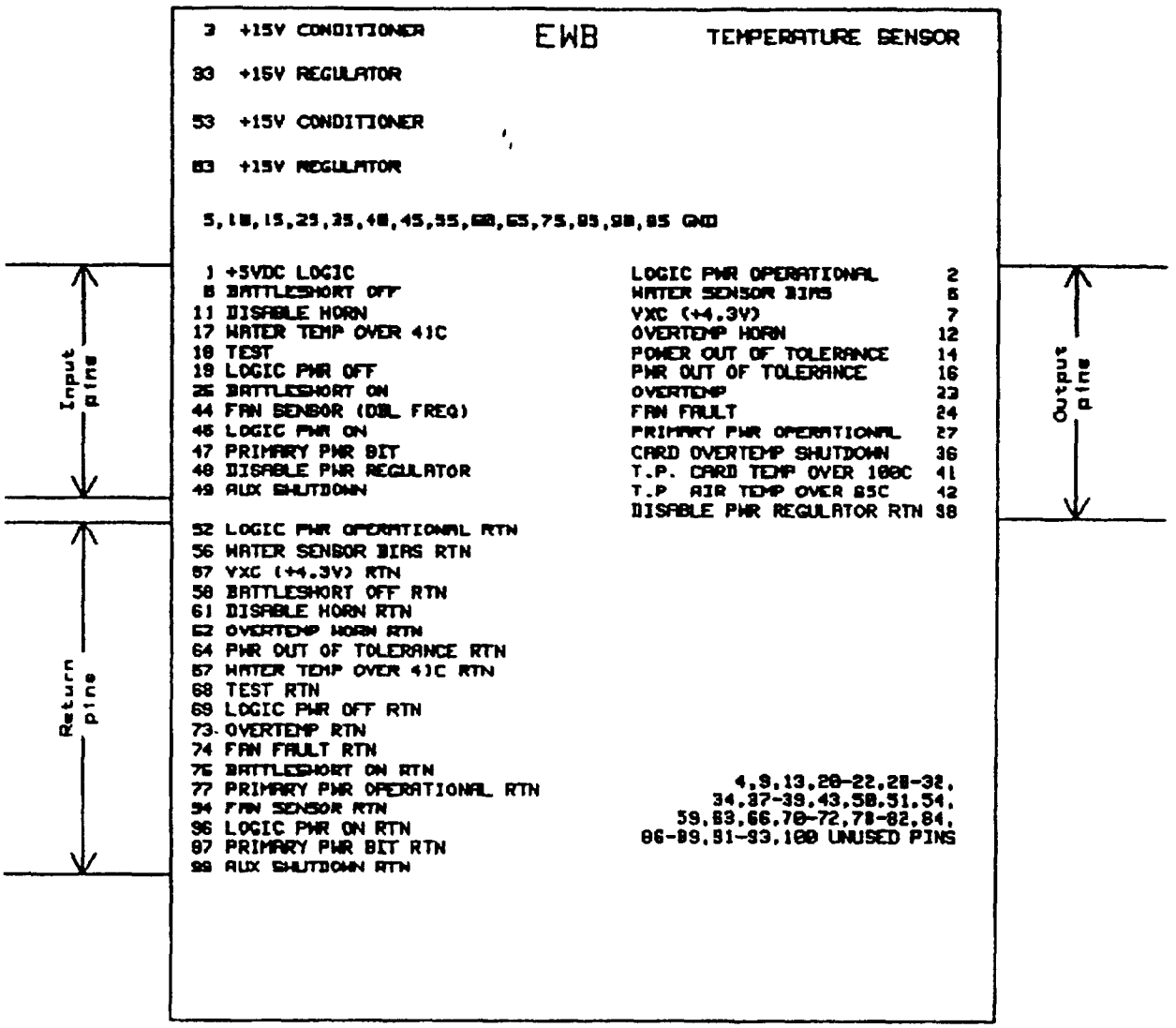


FIGURE 2. EWB pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, and E are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, C, D, and E where applicable, followed by the remaining tests in the sequence.

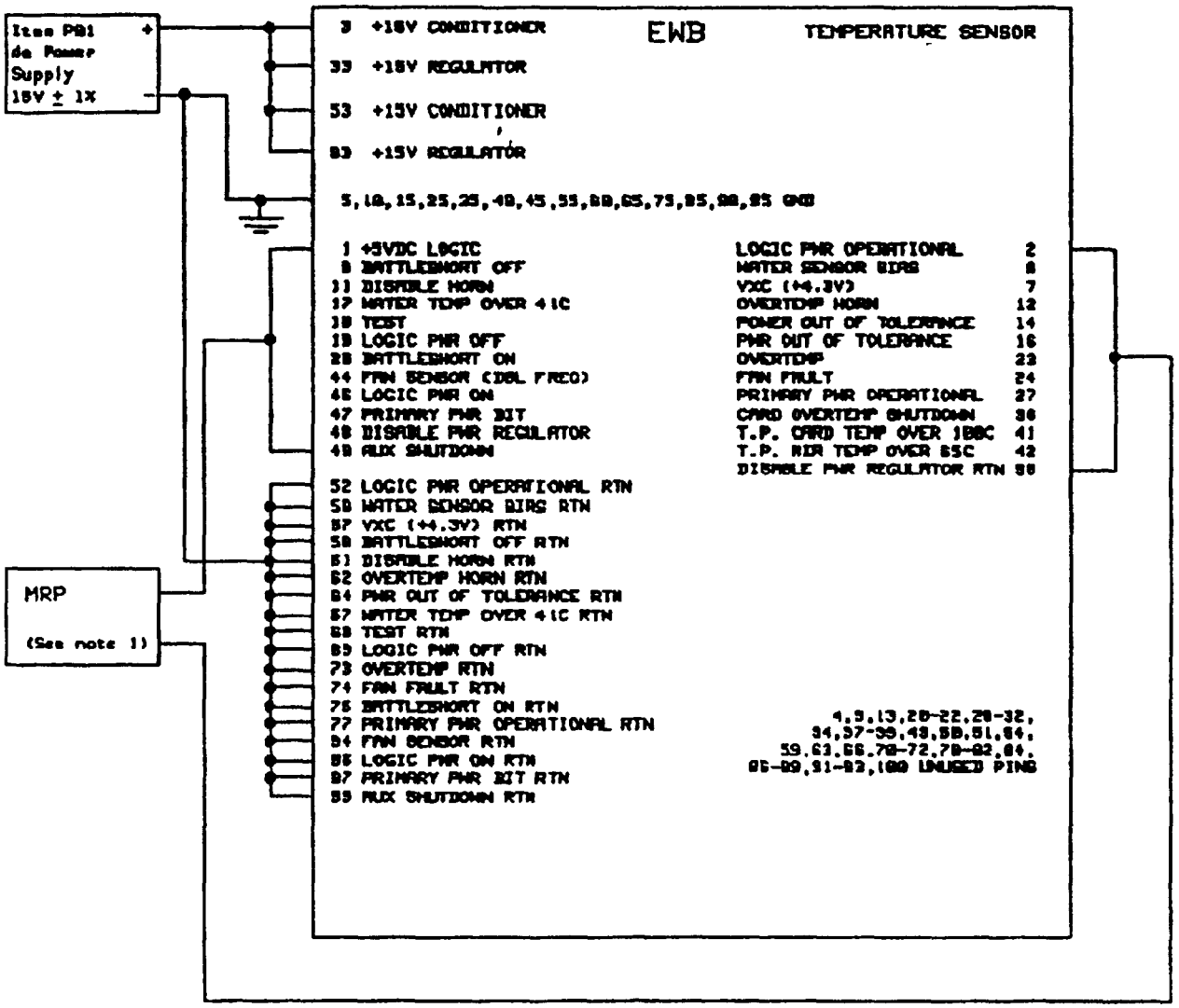
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. EWB operating temperature and component temperature test circuit.

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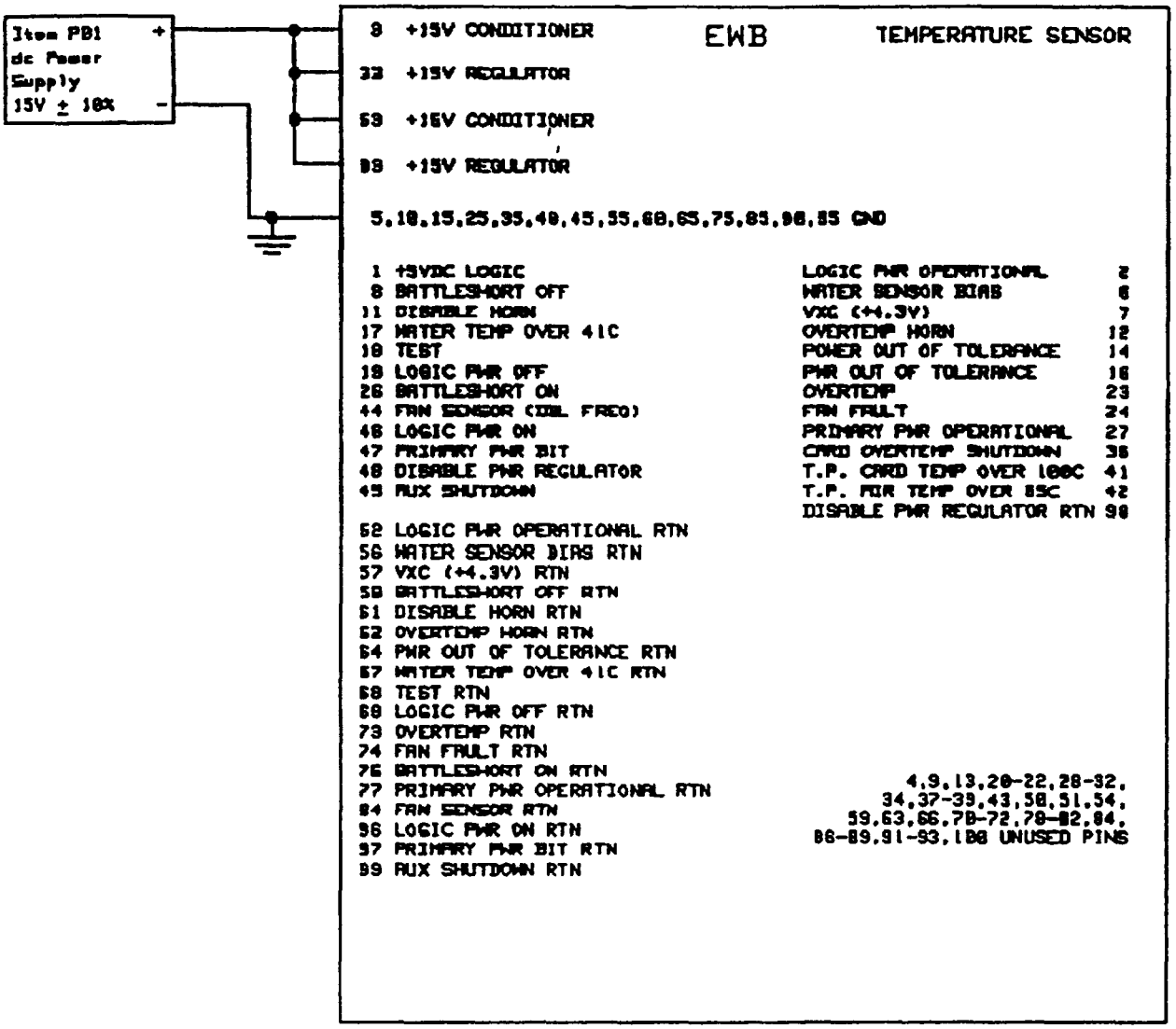


FIGURE 4. EWB vibration test circuit.

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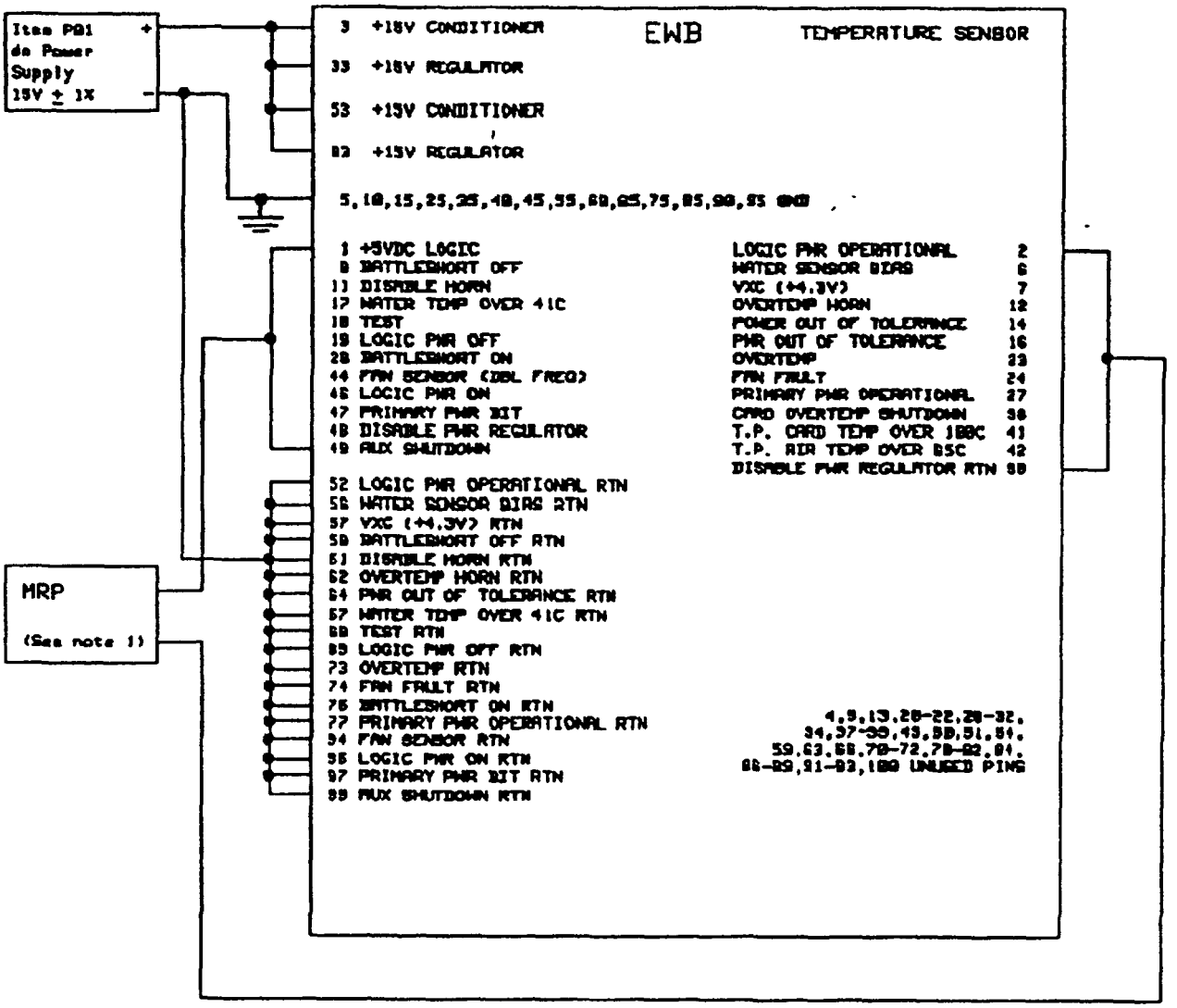
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. EWB life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vcc = power supply voltage.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 0.478 W (max); 0.32 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
+5VDC LOGIC	NA	NA	NA	NA	NA	NA	NA
BATTLE OFF	NA	0.4	2.4	-1.64	-0.46	0.339	75
DISABLE HORN	NA	0.4	2.4	-2.83	-0.44	NA	50
WATER TEMP	NA	0.4	2.4	-1.84	-0.46	0.339	75
TEST	NA	0.4	2.4	-2.24	-0.44	0.439	100
LOGIC OFF	NA	0.4	2.4	-1.84	-0.46	NA	75
BATTLE ON	NA	0.4	2.4	-1.84	-0.46	0.339	75
FAN SENSOR	NA	NA	NA	NA	NA	NA	50
LOGIC ON	NA	0.4	2.4	-1.64	-0.46	0.339	75
PRI PWR BIT	NA	NA	NA	NA	NA	NA	75
DISABLE PWR	NA	NA	NA	NA	NA	NA	100
AUX SHUTDOWN	NA	NA	NA	NA	NA	NA	125

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
LOGIC OPER	NA	NA	NA	NA	NA	75
WATER SENSOR	NA	NA	NA	NA	NA	100
VXC	NA	NA	NA	NA	NA	75
OVER T HORN	NA	0.5	-0.16	1.6	NA	50
PWR TOLER	NA	0.5	-0.16	1.6	NA	50
PWR TOLER	2.4	0.5	-0.4	6.0	-60.0	75
OVERTEMP	NA	0.5	-0.16	1.6	NA	75
FAN FAULT	NA	0.5	-0.16	1.6	NA	75
PRI PWR OPER	NA	0.5	-0.16	1.6	NA	75
CARD SH DOWN	2.4	NA	-0.4	6.0	-60.0	75
CARD TEMPOWN	NA	NA	NA	NA	NA	NA
AIR TEMP	NA	NA	NA	NA	NA	NA
DIS PWR RTN	NA	NA	NA	NA	NA	100

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6.3.3 Class I dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			0°C	25°C	60°C	0°C	25°C	60°C		
1	tpd	TEST to OVERTEMP	-	-	-	43	43	43	nS	11
2	tpd	PRI PWR BIT to PRI PWR OPER	-	-	-	80000	80000	80000	nS	11
3	tpd	WATER TEMP to OVER T HORN	-	-	-	86	86	86	nS	11
4	tpd	LOGIC OFF to PWR TOLER	-	-	-	43	43	43	nS	11
5	tpd	LOGIC OFF to DISABLE PWR	-	-	-	2800000	2800000	2800000	nS	11

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6.4 Module function. The temperature sensor module (EWB) contains the circuits required to advise the equipment operator when an overtemperature condition exists, when a fan fault occurs, if the primary power is operational, if the logic power is operational, and a power shutdown circuit that responds to DC power logic switch and to an overtemperature condition (if the BATTLESHORT ON/OFF switch is in OFF position). The power regulator will not turn on without the temperature sensor module in the Standard Electronic Module (SEM) chassis.

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6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Description
+5VDC LOGIC	1	I	+5VDC LOGIC
BATTLE OFF	8	I	BATTLESHORT OFF
DISABLE HORN	11	I	DISABLE HORN
WATER TEMP	17	I	WATER TEMP OVER 41C
TEST	18	I	TEST
LOGIC OFF	19	I	LOGIC PWR OFF
BATTLE ON	26	I	BATTLESHORT ON
FAN SENSOR	44	I	FAN SENSOR (DBL FREQ)
LOGIC ON	46	I	LOGIC PWR ON
PRI PWR BIT	47	I	PRIMARY PWR BIT
DISABLE PWR	48	I	DISABLE PWR REGULATOR
AUX SHUTDOWN	49	I	AUX SHUTDOWN
LOGIC OPER	2	O	LOGIC PWR OPERATIONAL
WATER SENSOR	6	O	WATER SENSOR BIAS
VXC	7	O	VXC (+4.3V)
OVER T HORN	12	O	OVERTEMP HORN
PWR TOLER	14	O	POWER OUT OF TOLERANCE
PWR TOLER	16	O	PWR OUT OF TOLERANCE
OVERTEMP	23	O	OVERTEMP
FAN FAULT	24	O	FAN FAULT

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
PRI PWR OPER	27	O	PRIMARY PWR OPERATIONAL
CARD SH DOWN	36	O	CARD OVERTEMP SHUTDOWN
CARD TEMPOWN	41	O	T.P. CARD TEMP OVER 100C
AIR TEMP	42	O	T.P. AIR TEMP OVER 65C
DIS PWR RTN	98	O	DISABLE PWR REGULATOR RTN

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0267)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,4,9, 13,20, 21,22, 28,29, 30,31, 32,34, 37,38, 39,43, 50,51, 54,59, 63,66, 70,71, 72,78, 79,80, 81,82, 84,86, 87,88, 89,91, 92,93, 100	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	4,9,13, 20,21, 22,28, 29,30, 31,32, 34,37, 38,39, 43,50, 51,54, 59,63, 66,70, 71,72, 78,79, 80,81, 82,84, 86,87, 88,89, 91,92, 93,100				
A3-40	Isolation, unused to unused pins	See note 4.		10	INF	MOhm	1-9

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A40
Isolation measurement	NA	± 500 KOhm	A1-A40

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 4,9,13,20,21,22,28,29, 30,31,32,34,37,38,39,43,50,51,54,59,63,66,70,71,72,78,79,80,81, 82,84,86,87,88,89,91,92,93,100.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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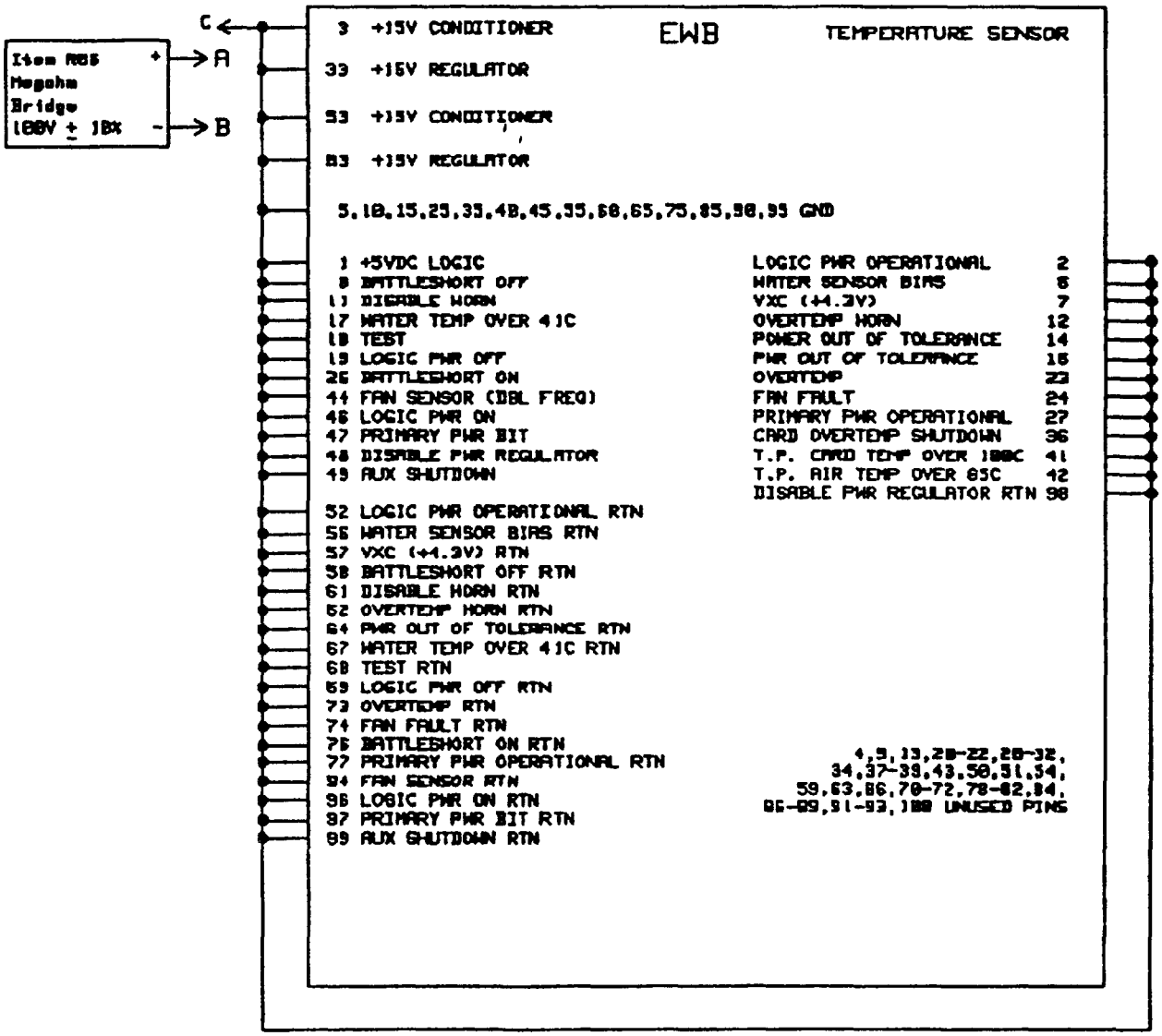


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, +15V Conditioner	3	53	0	1	Ohm	1-5
B2	Continuity, +15V Regulator	33	83				
B3	Continuity, GND	5	10				
B4	Continuity, GND	5	15				
B5	Continuity, GND	5	25				
B6	Continuity, GND	5	35				
B7	Continuity, GND	5	40				
B8	Continuity, GND	5	45				
B9	Continuity, GND	5	55				
B10	Continuity, GND	5	60				
B11	Continuity, GND	5	65				
B12	Continuity, GND	5	75				
B13	Continuity, GND	5	85				
B14	Continuity, GND	5	90				
B15	Continuity, GND	5	95				
B16	Continuity, RTN	52	5				
B17	Continuity, RTN	56	5				
B18	Continuity, RTN	57	5				
B19	Continuity, RTN	58	5	0	1	Ohm	1-5

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B20	Continuity, RTN	61	5	0	1	Ohm	1-5
B21	Continuity, RTN	62	5				
B22	Continuity, RTN	64	5				
B23	Continuity, RTN	67	5				
B24	Continuity, RTN	68	5				
B25	Continuity, RTN	69	5				
B26	Continuity, RTN	73	5				
B27	Continuity, RTN	74	5				
B28	Continuity, RTN	76	5				
B29	Continuity, RTN	77	5				
B30	Continuity, RTN	94	5				
B31	Continuity, RTN	96	5				
B32	Continuity, RTN	97	5				
B33	Continuity, RTN	99	5	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B33

2. The ohmmeter used shall be item R07.

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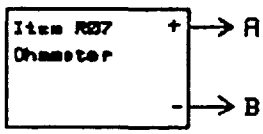
NOTES: - Continued.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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3	+15V CONDITIONER	EWB	TEMPERATURE SENSOR
33	+15V REGULATOR		
53	+15V CONDITIONER		
83	+15V REGULATOR		

5,10,15,25,35,40,45,55,60,65,75,85,90,95 GND

1	+5VDC LOGIC	LOGIC PWR OPERATIONAL	2
8	BATTLESHORT OFF	WATER SENSOR BIAS	6
11	DISABLE HORN	VXC (+4.3V)	7
17	WATER TEMP OVER 41C	OVERTEMP HORN	12
18	TEST	POWER OUT OF TOLERANCE	14
19	LOGIC PWR OFF	PWR OUT OF TOLERANCE	16
26	BATTLESHORT ON	OVERTEMP	23
44	FAN SENSOR (IBL FREQ)	FAN FAULT	24
46	LOGIC PWR ON	PRIMARY PWR OPERATIONAL	27
47	PRIMARY PWR BIT	CARD OVERTEMP SHUTDOWN	36
48	DISABLE PWR REGULATOR	T.P. CARD TEMP OVER 100C	41
49	AUX SHUTDOWN	T.P. AIR TEMP OVER 65C	42
		DISABLE PWR REGULATOR RTN	90
52	LOGIC PWR OPERATIONAL RTN		
56	WATER SENSOR BIAS RTN		
57	VXC (+4.3V) RTN		
58	BATTLESHORT OFF RTN		
61	DISABLE HORN RTN		
62	OVERTEMP HORN RTN		
64	PWR OUT OF TOLERANCE RTN		
67	WATER TEMP OVER 41C RTN		
68	TEST RTN		
69	LOGIC PWR OFF RTN		
73	OVERTEMP RTN		
74	FAN FAULT RTN		
76	BATTLESHORT ON RTN		
77	PRIMARY PWR OPERATIONAL RTN	1,9,13,20-22,28-32,	
94	FAN SENSOR RTN	34,37-39,43,50,51,54,	
96	LOGIC PWR ON RTN	59,63,66,70-72,78-82,84,	
97	PRIMARY PWR BIT RTN	86-89,91-93,100 UNUSED PINS	
99	AUX SHUTDOWN RTN		

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	17.0	3,33,53, 83	5,10,15, 25,35, 40,45, 55,60, 65,75, 85,90, 95	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	17.0 V	+0 mV,-170 mV	C
Power supply current limit	156.4 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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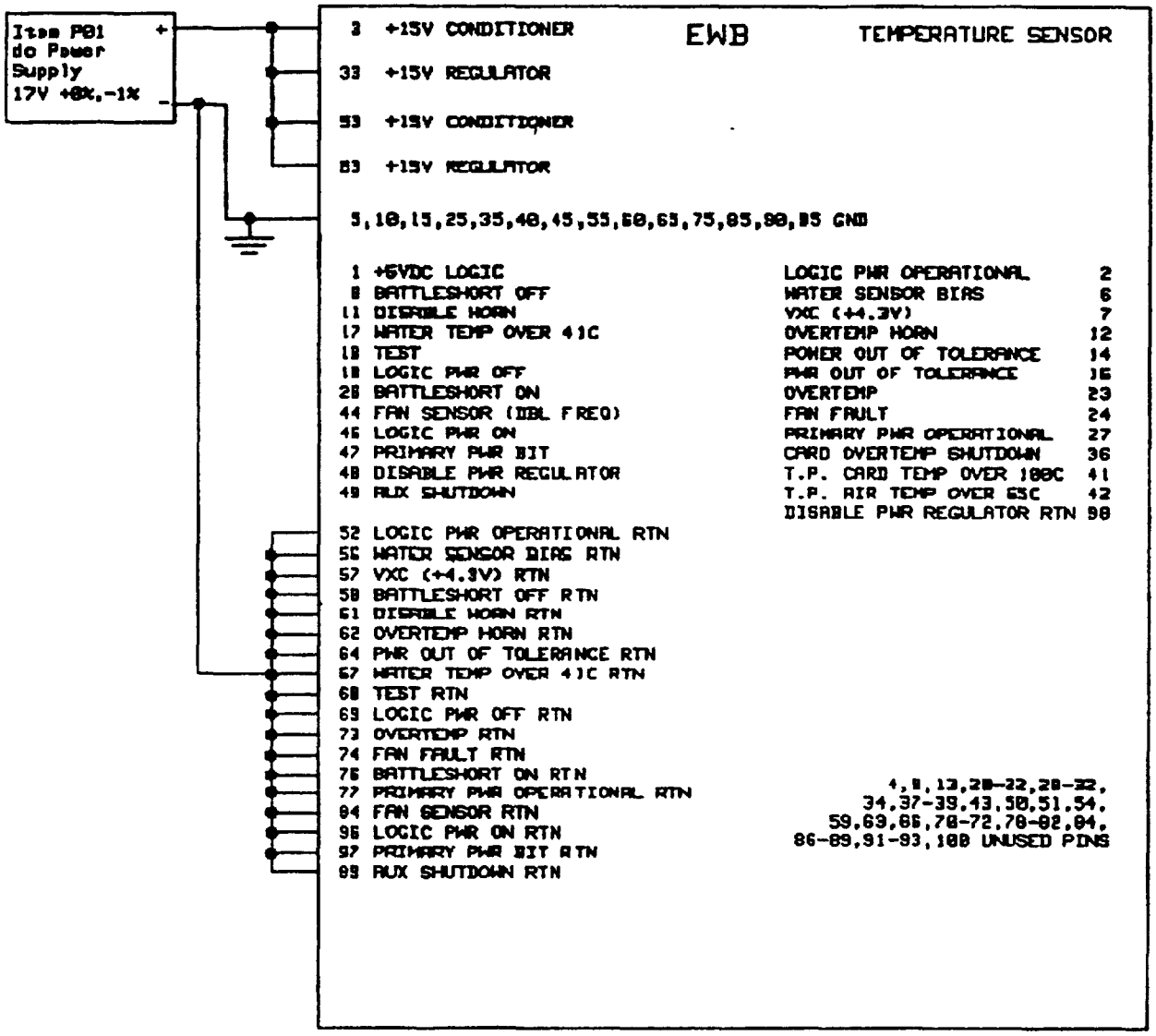


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	BATTLE OFF	8	5.5	-0.239	0.339	mA	1-3
D2	WATER TEMP	17	5.5	-0.239	0.339	mA	1-3
D3	TEST	18	5.5	-0.239	0.439	mA	1-3
D4	BATTLE ON	26	5.5	-0.239	0.339	mA	1-3
D5	LOGIC ON	46	5.5	-0.239	0.339	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D5
Power supply current limit	1130 mA	NA	D1-D5
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D5
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D5 D1-D5
Input current measurement	 NA NA NA	 ± 28.9 uA ± 33.9 uA ± 28.9 uA	 D1,D2 D3 D4,D5

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	PWR TOLER	16	0.0	-150.0	-60.0	mA	1-4
E2	CARD SH DOWN	36	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1,E2
Power supply current limit	1130 mA	NA	E1,E2
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1,E2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1,E2 E1,E2
Output current measurement	NA	± 4.5 mA	E1,E2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F1	BATTLE OFF	8	2.4	-1.64	-0.46	mA	1-3
F2	DISABLE HORN	11	2.4	-2.83	-0.44	mA	1-3
F3	WATER TEMP	17	2.4	-1.84	-0.46	mA	1-3
F4	TEST	18	2.4	-2.24	-0.44	mA	1-3
F5	LOGIC OFF	19	2.4	-1.84	-0.46	mA	1-3
F6	BATTLE ON	26	2.4	-1.84	-0.46	mA	1-3
F7	LOGIC ON	46	2.4	-1.64	-0.46	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F7
Power supply current limit	1130 mA	NA	F1-F7
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F7
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F7 F1-F7
Input current measurement	NA NA NA NA NA NA	± 59 uA ± 119 uA ± 69 uA ± 90 uA ± 69 uA ± 59 uA	F1 F2 F3 F4 F5,F6 F7

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	BATTLE OFF	8	0.4	-1.64	-0.46	mA	1-3
G2	DISABLE HORN	11	0.4	-2.83	-0.44	mA	1-3
G3	WATER TEMP	17	0.4	-1.84	-0.46	mA	1-3
G4	TEST	18	0.4	-2.24	-0.44	mA	1-3
G5	LOGIC OFF	19	0.4	-1.84	-0.46	mA	1-3
G6	BATTLE ON	26	0.4	-1.84	-0.46	mA	1-3
G7	LOGIC ON	46	0.4	-1.64	-0.46	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G7
Power supply current limit	1130 mA	NA	G1-G7
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G7
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G7 G1-G7
Input current measurement	 NA NA NA NA NA NA	 ± 59 uA ± 119 uA ± 69 uA ± 90 uA ± 69 uA ± 59 uA	 G1 G2 G3 G4 G5,G6 G7

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	PWR TOLER	16	-0.4	2.4	4.5	V	1-4
H2	CARD SH DOWN	36	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1,H2
Power supply current limit	1130 mA	NA	H1,H2
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1,H2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1,H2 H1,H2
Output voltage measurement	NA	± 105 mV	H1,H2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

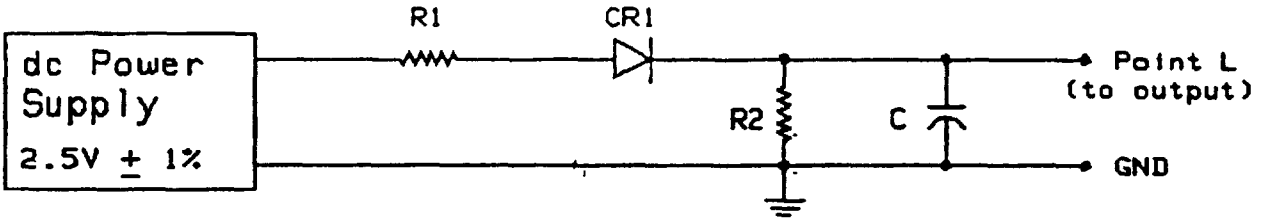
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
16	6200	6.0	0.5
12,14,23,24,27	16000	1.6	0.5
36	6200	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	OVER T HORN	12	1.6	0.0	0.5	V	1-4
J2	PWR TOLER	14	1.6				
J3	PWR TOLER	16	6.0				
J4	OVERTEMP	23	1.6				
J5	FAN FAULT	24					
J6	PRI PWR OPER	27	1.6	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J6
Power supply current limit	1130 mA	NA	J1-J6
Forcing current, (Iol)	1.6 mA 6.0 mA 1.6 mA	± 160 uA ± 600 uA ± 160 uA	J1,J2 J3 J4-J6
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J6 J1-J6
Output voltage measurement	NA	± 25 mV	J1-J6

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	33	2.1	3.9	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 90 nF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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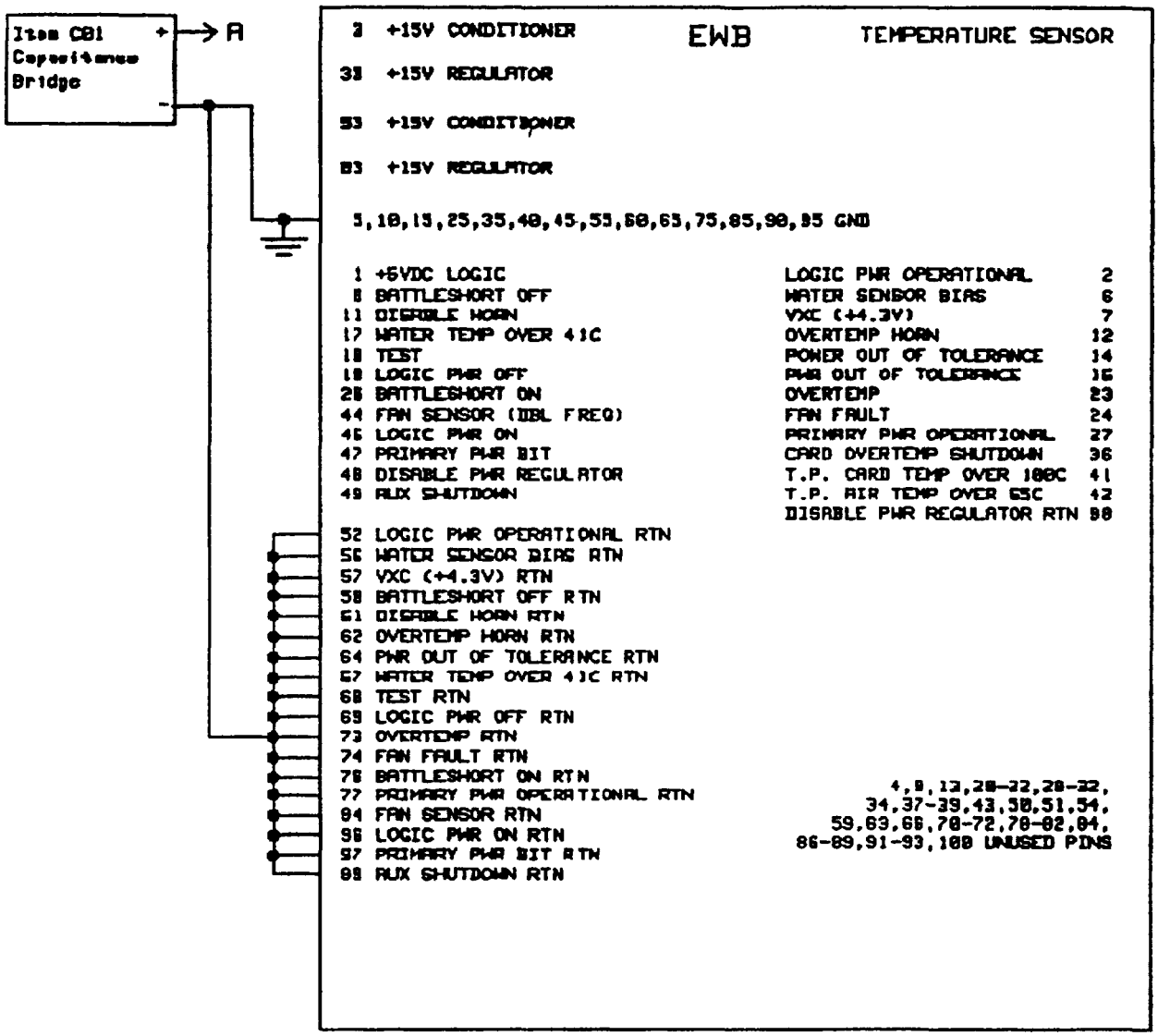


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
N	Icc	33	36.0	86.9	mA	1-3

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NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1130 mA	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 2.54 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1130 mA	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1130 mA	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

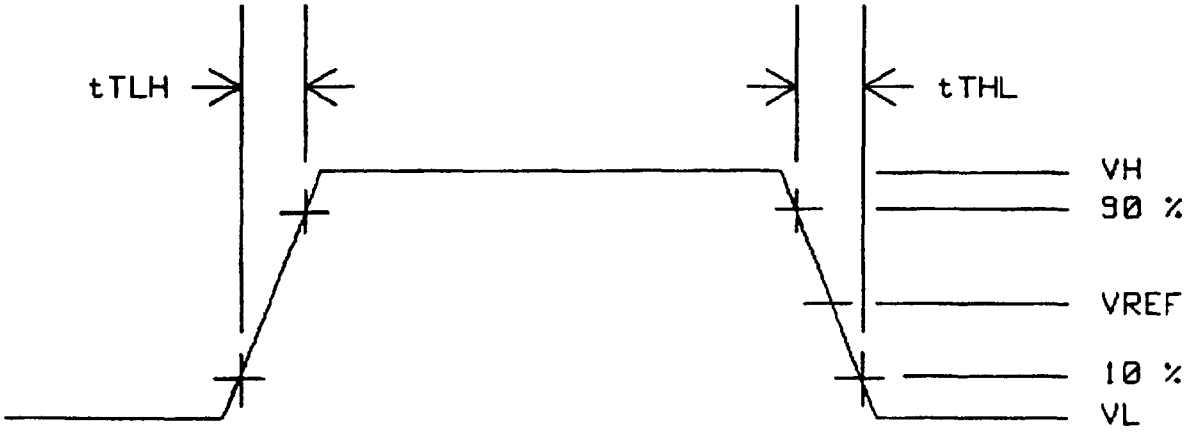
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XV. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			0°C	25°C	60°C	0°C	25°C	60°C		
1	tpd	TEST to OVERTEMP	-	-	-	43	43	43	nS	11
2	tpd	PRI PWR BIT to PRI PWR OPER	-	-	-	80000	80000	80000	nS	11
3	tpd	WATER TEMP to OVER T HORN	-	-	-	86	86	86	nS	11
4	tpd	LOGIC OFF to PWR TOLER	-	-	-	43	43	43	nS	11
5	tpd	LOGIC OFF to DISABLE PWR	-	-	-	2800000	2800000	2800000	nS	11

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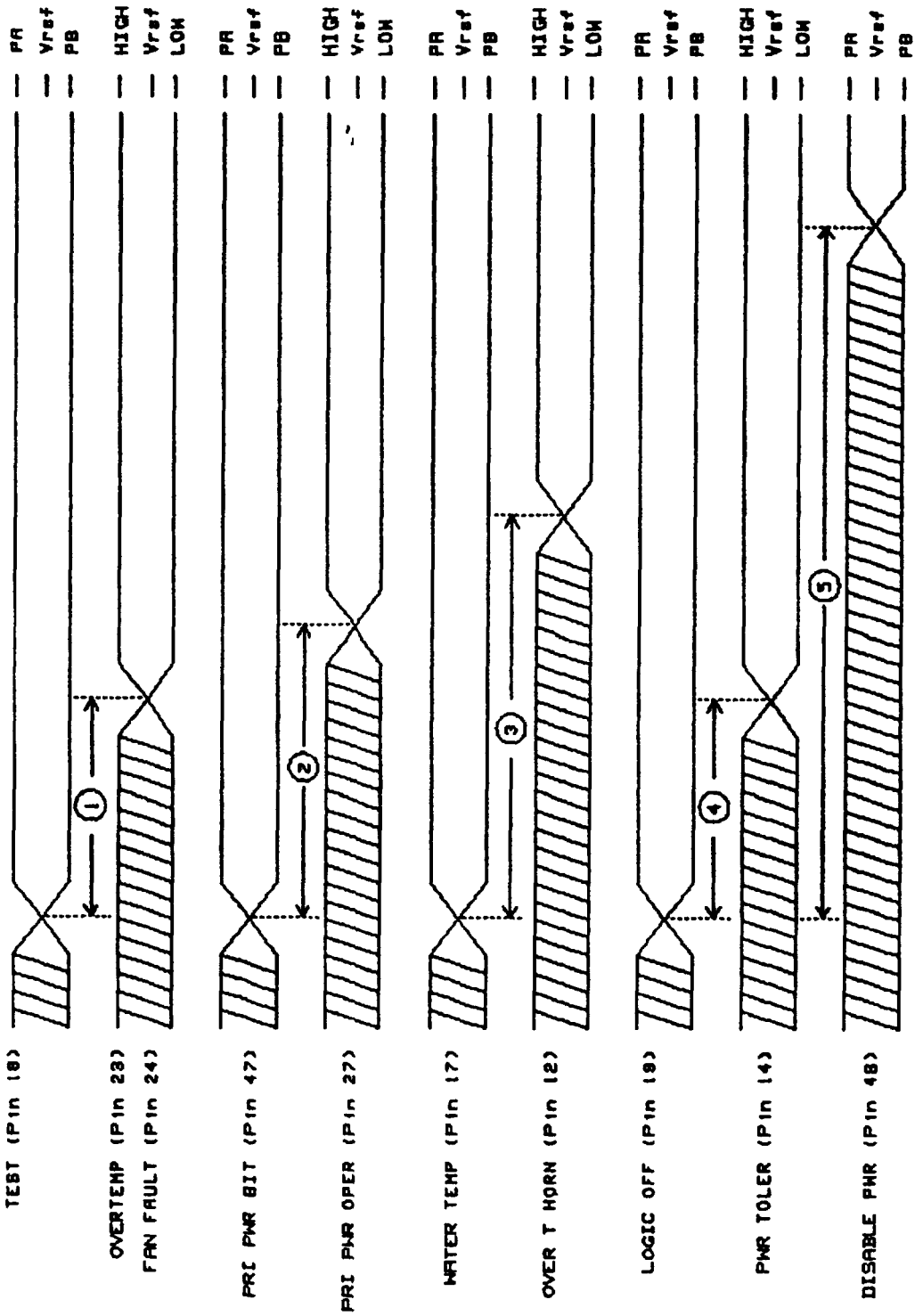


FIGURE 11. Dynamic waveforms for table XVI.

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TABLE XVI. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	BATTLE OFF	8	2	75	pF	1-4
R2	DISABLE HORN	11	2	50	pF	1-4
R3	WATER TEMP	17	2	75	pF	1-4
R4	TEST	18	2	100	pF	1-4
R5	LOGIC OFF	19	2	75	pF	1-4
R6	BATTLE ON	26	2	75	pF	1-4
R7	FAN SENSOR	44	2	50	pF	1-4
Outputs						
R8	PWR TOLER	16	8	75	pF	1-4
R9	CARD SH DOWN	36	6	75	pF	1-4
Inputs						
R10	LOGIC ON	46	4	75	pF	1-4
R11	PRI PWR BIT	47	2	75	pF	1-4
R12	DISABLE PWR	48	2	100	pF	1-4
R13	AUX SHUTDOWN	49	2	125	pF	1-4
Outputs						
R14	LOGIC OPER	2	2	75	pF	1-4
R15	WATER SENSOR	6	2	100	pF	1-4
R16	VXC	7	2	75	pF	1-4
R17	OVER T HORN	12	2	50	pF	1-4
R18	PWR TOLER	14	2	50	pF	1-4
R19	OVERTEMP	23	2	75	pF	1-4

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TABLE XVI. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R20	FAN FAULT	24	2	75	pF	1-4
R21	PRI PWR OPER	27	2	75	pF	1-4
R22	DIS PWR RTN	98	2	100	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R19
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R19
Capacitance measurement	NA	± 3.65 pF	R1
	NA	± 2.4 pF	R2
	NA	± 3.65 pF	R3
	NA	± 4.9 pF	R4
	NA	± 3.65 pF	R5,R6
	NA	± 2.4 pF	R7
	NA	± 3.35 pF	R8
	NA	± 3.45 pF	R9
	NA	± 3.55 pF	R10
	NA	± 3.65 pF	R11
	NA	± 4.9 pF	R12
	NA	± 6.15 pF	R13
	NA	± 3.65 pF	R14
	NA	± 4.9 pF	R15
	NA	± 3.65 pF	R16
	NA	± 2.4 pF	R17,R18
	NA	± 3.65 pF	R19-R21
	NA	± 4.9 pF	R22

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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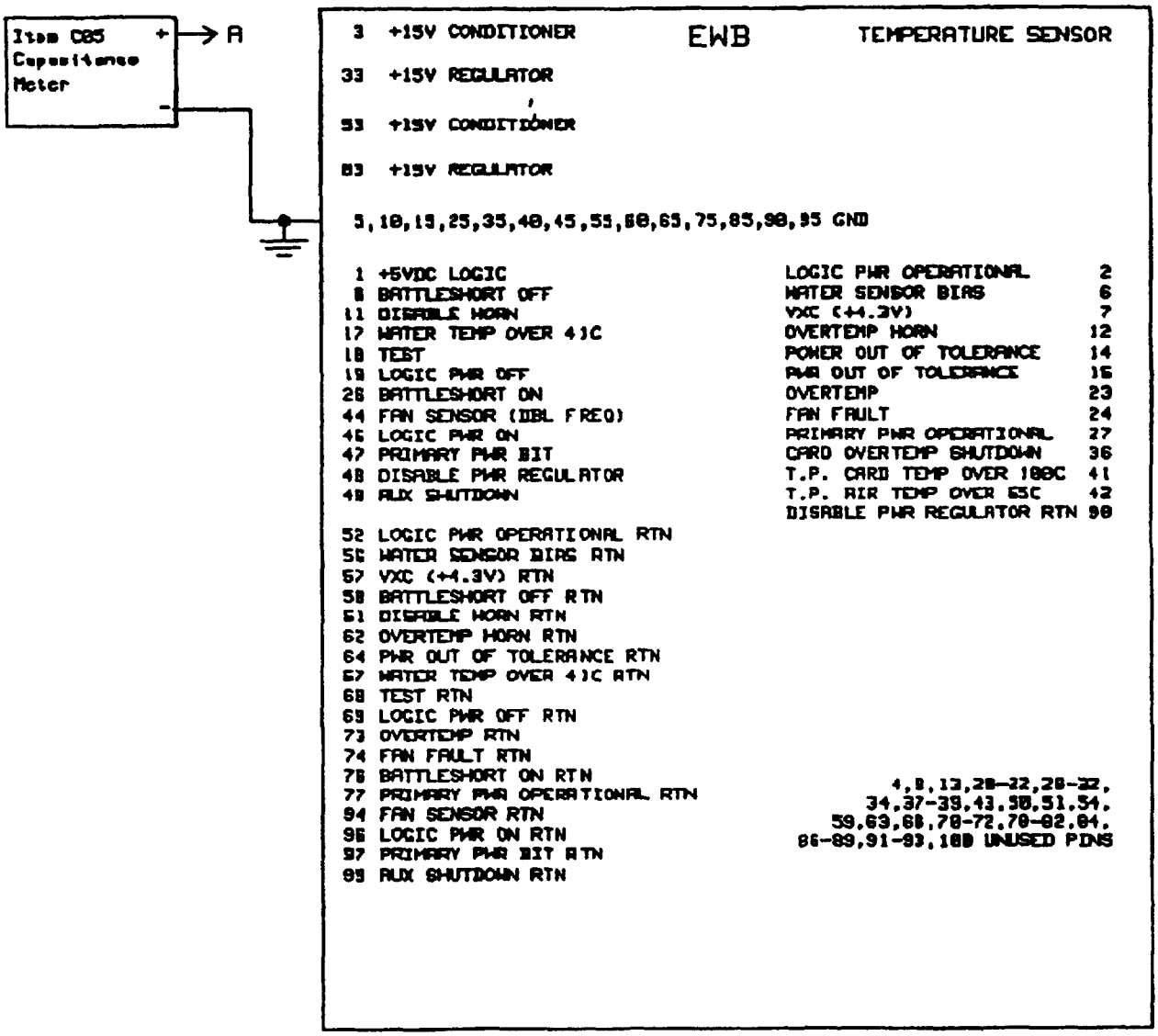


FIGURE 12. Input and output capacitance test circuit for table XVII.

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TABLE XVII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	3	0	400	mV	1-4
T2	PSTA	53	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1130 mA	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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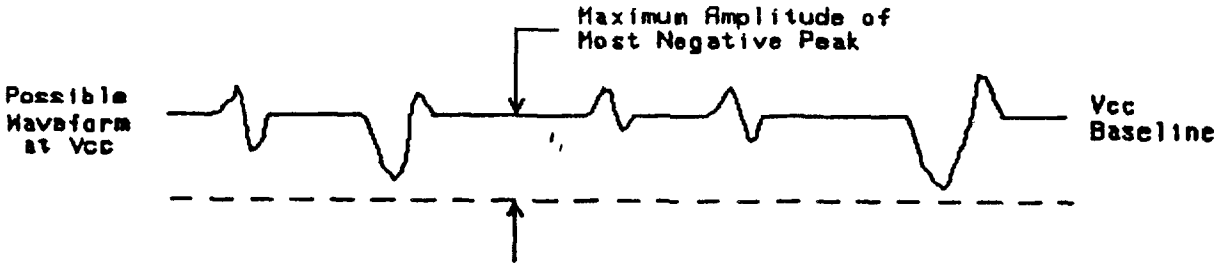
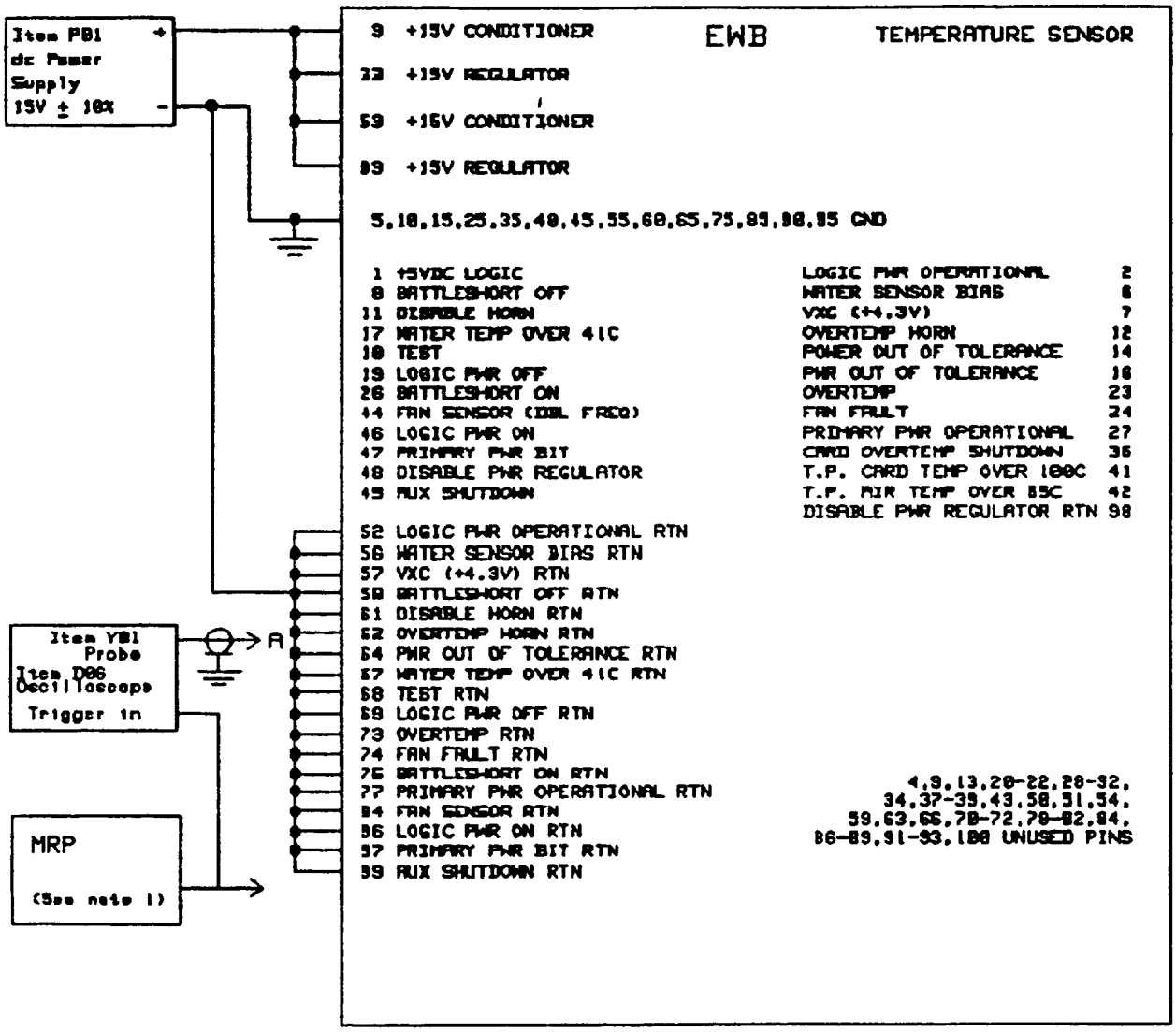


FIGURE 13. Power supply transient amplitude waveform for table XVIII.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XVIII.

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